No Uniform Finite Family of Modularity Testers for Finite Partition Lattices

Automated mathematics run `fa_banach_001`, lane 1

22 July 2026

Status. Counterexample / full negative answer to Question 6.47 of Contreras Mantilla–Sinclair [1]. Verdict: likely valid, pending human review.

1 Source question

Let P_n be the lattice of partitions of an n-element set, ordered from the discrete partition 0 to the indiscrete partition 1. Write $\#x$ for the number of blocks and

$$|x| = \frac{n - \#x}{n-1}, \qquad d(x,y) = 2|x \vee y| - |x| - |y|.$$

A partition is *singular* when it has at most one non-singleton block; the singular set is denoted Σ_n. The source uses

$$\varphi(x,y) = \inf_z \max\left\{(|x| + |y|) \mathbin{\dot{-}} (|x \vee y| + |z|), d(x \vee z, x), d(y \vee z, y)\right\},$$

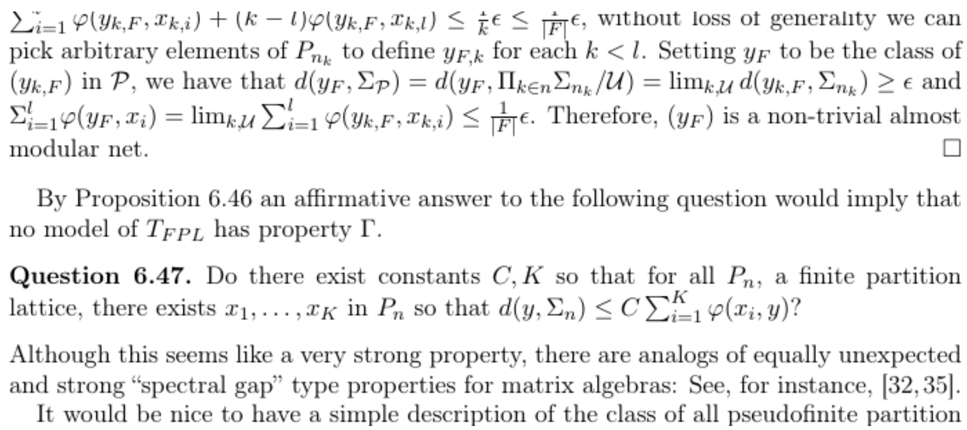
where $a \mathbin{\dot{-}} b = \max\{a - b, 0\}$. In a complete metric lattice, $\varphi(x,y) = 0$ exactly when (x,y) is a metrically modular pair [1, Remark following (15)].

Question 6.47 on page 51 asks:

> Do there exist constants C, K so that for all P_n, a finite partition lattice, there exists $x_1, \ldots, x_K \in P_n$ so that
> $$d(y, \Sigma_n) \leq C \sum_{i=1}^{K} \varphi(x_i, y)?$$

2 Proof intuition

A fixed tester partition only records, for any pair of points, whether they belong to the same block. Thus K testers assign one of only 2^K colors to each unordered pair. Ramsey's theorem supplies four points whose six pairs all have the same color. Coordinate by coordinate, the four points are then either all in one tester block or all in distinct tester blocks. Pairing the four points in a new partition creates two non-singleton blocks, yet in the first case this new partition refines the tester and in the second case the two pair-merges are rank-independent. Both cases give exact modularity.



$\sum_{i=1}^{l} \varphi(y_{k,F}, x_{k,i}) + (k-l)\varphi(y_{k,F}, x_{k,l}) \le \frac{1}{k}\epsilon \le \frac{1}{|F|}\epsilon$, without loss of generality we can pick arbitrary elements of P_{n_k} to define $y_{F,k}$ for each $k < l$. Setting y_F to be the class of $(y_{k,F})$ in $\mathcal{P}$, we have that $d(y_F, \Sigma_{\mathcal{P}}) = d(y_F, \Pi_{k\in n}\Sigma_{n_k}/\mathcal{U}) = \lim_{k,\mathcal{U}} d(y_{k,F}, \Sigma_{n_k}) \ge \epsilon$ and $\Sigma_{i=1}^{l}\varphi(y_F, x_i) = \lim_{k,\mathcal{U}} \sum_{i=1}^{l} \varphi(y_{k,F}, x_{k,i}) \le \frac{1}{|F|}\epsilon$. Therefore, (y_F) is a non-trivial almost modular net. □

By Proposition 6.46 an affirmative answer to the following question would imply that no model of T_{FPL} has property Γ.

Question 6.47. Do there exist constants C, K so that for all P_n, a finite partition lattice, there exists $x_1, \ldots, x_K$ in P_n so that $d(y, \Sigma_n) \le C \sum_{i=1}^{K} \varphi(x_i, y)$?

Although this seems like a very strong property, there are analogs of equally unexpected and strong "spectral gap" type properties for matrix algebras: See, for instance, [32, 35].

It would be nice to have a simple description of the class of all pseudofinite partition

Figure 1: Question 6.47 in the source paper, PDF page 51.

3 Negative answer

Let $R_r(4)$ denote a finite Ramsey number: every r-coloring of the pairs of a set of size at least $R_r(4)$ has a monochromatic four-element subset [2].

Theorem 1. *For every integer $K \ge 1$, every*

$$n \ge R_{2^K}(4),$$

and every choice $x_1, \ldots, x_K \in P_n$, there is a partition $y \in P_n$ with exactly two non-singleton blocks such that

$$\varphi(x_i, y) = 0 \quad (1 \le i \le K), \qquad d(y, \Sigma_n) = \frac{1}{n-1}.$$

Consequently the answer to Question 6.47 is negative.

Proof. Let V be the n-element ground set. Color every pair $\{u, v\} \in \binom{V}{2}$ by

$$c(\{u, v\}) = \left(\mathbf{1}_{u \sim_{x_1} v}, \ldots, \mathbf{1}_{u \sim_{x_K} v}\right) \in \{0,1\}^K,$$

where $u \sim_x v$ means that u, v lie in the same block of x. This uses at most 2^K colors. Ramsey's theorem gives $H = \{a, b, c, d\} \subseteq V$ whose six pairs have one common color.

Fix i. If the ith coordinate of the common color is 1, transitivity of $\sim_{x_i}$ says that all four points of H lie in one block of x_i. If it is 0, the six pair relations all fail, so the four points lie in four distinct blocks of x_i.

Define y to have the two blocks $\{a, b\}$ and $\{c, d\}$, with every other point a singleton. We show that (x_i, y) is metrically modular. In the first case above, $y \le x_i$, hence

$$x_i \vee y = x_i, \qquad x_i \wedge y = y,$$

and the modular rank identity is immediate. In the second case, $x_i \wedge y = 0$, while joining with y merges two disjoint pairs of four distinct x_i-blocks. Therefore

$$\#(x_i \vee y) = \#x_i - 2, \qquad |x_i \vee y| = |x_i| + \frac{2}{n-1} = |x_i| + |y|.$$

Again $|x_i \vee y| + |x_i \wedge y| = |x_i| + |y|$. Thus (x_i, y) is metrically modular in either case, and hence $\varphi(x_i, y) = 0$.

The partition y has two non-singleton blocks. Lemma 6.3 of the source gives

$$d(y, \Sigma_n) = \frac{[y] - 1}{n - 1} = \frac{1}{n - 1} > 0,$$

where $[y]$ is the number of non-singleton blocks.

If constants C, K as in Question 6.47 existed, choose $n \geq R_{2^K}(4)$ and apply the preceding construction to the proposed testers. Its right-hand side is zero and its left-hand side is positive, a contradiction. $\square$

4 Verification notes

The proof uses only three checkable inputs.

1. The pair-color has at most 2^K values, so finite Ramsey applies.
2. On a monochromatic four-set, each tester restricts either to one block or to four singleton blocks; there is no intermediate pattern.
3. In the four-singleton case, the two blocks of y merge four distinct tester blocks in two disconnected pairs, so the tester's block count drops by exactly two.

The script `code/verify_ramsey_witness.py` checks the two rank cases for concrete inputs. A separate exact enumeration of φ for $n \leq 6$ is included as `code/compute_small_gamma.py`; this computation motivated the argument but is not a dependency of the proof.

5 Scope and novelty check

This theorem completely answers Question 6.47, but it does not answer Questions 6.44 or 6.45 about property Γ. Indeed, the constructed witness is only $1/(n-1)$ away from Σ_n; the fixed positive-distance condition in Proposition 6.46 is therefore untouched.

A bounded novelty check was performed on 22 July 2026: the run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2507.10932 and the core terms; the locally parsed arXiv corpus was searched for the exact spectral-gap question and close phrases; and arXiv web searches were run for the exact question, the paper title with property Γ, and metric-lattice spectral-gap terms. No later answer or close Ramsey formulation was found. The arXiv record still lists only the 15 July 2025 version, in which Question 6.47 remains present. This is a bounded search, not a claim of exhaustive novelty.

6 Human-review recommendation

High priority. A reviewer should verify the order convention in the comparable case and the two-block-count drop in the four-distinct-block case. Both are explicit above; no asymptotic estimate or computational lemma is used.

References

[1] J. Contreras Mantilla and T. Sinclair, *The model theory of metric lattices: pseudofinite partition lattices*, arXiv:2507.10932, 2025.

[2] R. L. Graham, B. L. Rothschild, and J. H. Spencer, *Ramsey Theory*, 2nd ed., Wiley, 1990.